Optimal control

Model answers

Question 1

a) The Hamiltonian is

$$H(x(t), u(t), p(t)) = p(t)u(t) - \left(x(t) - t^2\right)^2$$

in which it is assumed the costate has been normalised so that $p_0 = -1$. The switching function is $\varphi(t) = p^*(t)$ associated with a specific optimal trajectory.

[3 marks]

b) The adjoint equation is

$$\dot{p}^*(t) = -H_x|_* = 2\left(x^*(t) - t^2\right) \tag{1}$$

The boundary conditions are $x^*(0) = 1$ and $p^*(2) = 0$.

[3 marks]

c) For a singular optimal control to exist, $\varphi(t)$ must identically vanish ($\varphi(t) \equiv 0$) on some time interval $t \in [t_1, t_2]$, that is if there is a time interval $[t_1, t_2]$ during which $\varphi(t) = 0$ for all $t \in [t_1, t_2]$ then $[t_1, t_2]$ is a singular interval. We investigate this possibility by noting that if $\varphi(t) = 0$ for $t \in [t_1, t_2]$ then

$$\varphi(t) = \dot{\varphi}(t) = \ddot{\varphi}(t) = \ldots = 0, \qquad t \in [t_1, t_2]$$

Answers

During a singular interval

$$\varphi(t) = p^*(t) = 0 \tag{2}$$

and by differentiating (2) and using (1)

$$\dot{\varphi}(t) = \dot{p}^*(t) = 2\left(x^*(t) - t^2\right) = 0 \tag{3}$$

which means

$$x^*(t) = t^2$$

By considering the state equation, it can be seen that for the control

$$u^*(t) = \dot{x}^*(t) = 2t \tag{4}$$

$\dot{\varphi}(t) = 0$ and all higher-order derivatives of the switching function are obviously 0, hence, $\varphi(t) \equiv 0$ and therefore singular arcs exist. The singular control law is given by (4). By considering the maximisation of the Hamiltonian and the singular control law, the optimal control in terms of the switching function is given by

$$u^*(t) = \begin{cases} 4 & \text{if } p(t) > 0 \\ 2t & \text{if } p(t) = 0 \\ 0 & \text{if } p(t) < 0 \end{cases}$$

[6 marks]

d) Since $x^*(t) \geq t^2$ for $t \in [0, 2]$, from (1)

$$\dot{p}^*(t) \geq 0 \text{ for } t \in [0, 2] \tag{5}$$

But the boundary condition $p^*(2) = 0$ and (5) imply that

$$p^*(t) \leq 0 \text{ for } t \in [0, 2]$$

Therefore the solution of $p^*(t)$ involves a "bang off" segment

$$p^*(t) < 0 \text{ for } t \in [0, t_k)$$

and a singular arc

$$p^*(t) = 0 \text{ for } t \in [t_k, 2]$$

where t_k is some time point in the interval $[0, 2]$. Note that, the cases of $t_k = 0$ and $t_k = 2$ can be shown to lead to a contradiction and are excluded. The optimal control is therefore

$$u^*(t) = \begin{cases} 0 & \text{for } 0 \leq t < t_k \\ 2t & \text{for } t_k \leq t < 2 \end{cases}$$

By applying this control to the state equation, gives the optimal state trajectory as

$$x^*(t) = \begin{cases} 1 & \text{for } 0 \leq t < t_k \\ t^2 + (1 - t_k^2) & \text{for } t_k \leq t < 2 \end{cases} \tag{6}$$

To calculate t_k, recall that on the singular arc $p^*(t) = \dot{p}^*(t) = 0$ for $t \in [t_k, 2]$, and by using (6) and (1), yields

$$2\left(1 - t_k^2\right) = 0$$

which gives

$$t_k = 1$$

i) Hence the optimal control is

$$u^*(t) = \begin{cases} 0 & \text{for } 0 \leq t < 1 \\ 2t & \text{for } 1 \leq t < 2 \end{cases}$$

[4 marks]

ii) The optimal state trajectory is

$$x^*(t) = \begin{cases} 1 & \text{for } 0 \leq t < 1 \\ t^2 & \text{for } 1 \leq t < 2 \end{cases}$$

[4 marks]

Answers

Question 2

a) The state variables are the position coordinates $x(t)$ and $y(t)$ $\left(\boldsymbol{x}(t) = [x(t)\ y(t)]^T\right)$, and the control variable is the heading angle $u(t) = \psi(t)$.

[2 marks]

b) The functional to be minimised is

$$J(u) = \int_0^2 -y(t)\ dt.$$

[2 marks]

c) The Hamiltonian is

$$H(\boldsymbol{x}(t), u(t), p(t)) = p_1(t)V\cos\psi(t) + p_2(t)V\sin\psi(t) + y(t)$$

in which it is assumed that $p(t)$ has been normalised so that $p_0 = -1$.

[2 marks]

d) The control variable $\psi(t)$ must be selected to maximise

$$H(\boldsymbol{x}(t), u(t), p(t)) = p_1(t)V\cos\psi(t) + p_2(t)V\sin\psi(t) + y(t)$$

which is equivalent to maximising

$$p_1(t)\cos\psi(t) + p_2(t)\sin\psi(t) = \sqrt{p_1^2(t) + p_2^2(t)}\,\sin(\psi(t) + \beta(t)) \tag{7}$$

in which $\beta(t)$ satisfies

$$\sin\beta(t) = \frac{p_1(t)}{\sqrt{p_1^2(t) + p_2^2(t)}} \qquad \text{and} \qquad \cos\beta(t) = \frac{p_2(t)}{\sqrt{p_1^2(t) + p_2^2(t)}}$$

The Hamiltonian is therefore maximised when

$$\psi^*(t) + \beta^*(t) = \frac{\pi}{2}$$

and the optimal control is

$$\psi^*(t) = -\beta^*(t) + \frac{\pi}{2} \tag{8}$$

By taking the tan on both sides, the optimal control satisfies

$$\tan\psi^*(t) = \tan\left(-\beta^*(t) + \frac{\pi}{2}\right) = \frac{1}{\tan\beta^*(t)} = \frac{\cos\beta^*(t)}{\sin\beta^*(t)} = \frac{p_2^*(t)}{p_1^*(t)}$$

[3 marks]

e) The adjoint equations are

$$\begin{aligned} \dot{p}_1^*(t) &= -H_x|_* = 0 \\ \dot{p}_2^*(t) &= -H_y|_* = -1 \end{aligned} \tag{9}$$

[2 marks]

Answers

f) The equations in (9) give

$$p_1^*(t) = A \qquad \text{and} \qquad p_2^*(t) = -t + B$$

where A and B are two constants of integration. The optimal control is therefore given by

$$\tan\psi^*(t) = \frac{p_2^*(t)}{p_1^*(t)} = \frac{-t+B}{A} = a_1 t + a_2$$

where $a_1 \,(= -\frac{1}{A})$ and $a_2 \,(= \frac{B}{A})$ are two constants.

[2 marks]

g) By using the substitution $z = a_1 t + a_2$

$$\psi^*(t) = \arctan(z)$$

Then by substitution into the state equations, yields

$$\dot{x}^*(t) = V\cos(\arctan z) = \frac{V}{\sqrt{1+z^2}}$$

and

$$\dot{y}^*(t) = V\sin(\arctan z) = \frac{Vz}{\sqrt{1+z^2}}$$

Then by integration of the above two equations, respectively

$$x^*(t) = \frac{V}{a_1}\ln\left|z + \sqrt{z^2+1}\right| + C_1 \tag{10}$$

and

$$y^*(t) = \frac{V}{a_1}\sqrt{1+z^2} + C_2 \tag{11}$$

where C_1 and C_2 are two constants of integration. By rearranging (11)

$$z = \pm\sqrt{\frac{a_1^2}{V^2}\left(y^*(t) - C_2\right)^2 - 1} \tag{12}$$

Then by substituting (12) into (10) and after a few calculation steps, the extremal curve followed by the boat is

$$y(x) = -\frac{V}{a_1}\cosh\left(\frac{a_1}{V}\left(x - C_1\right)\right) + C_2$$

[7 marks]

Answers

Question 3

a) The minimisation problem is

$$\min_{u(t)} \int_0^2 -y(t)\, dt = \min_{u(t)} \int_0^2 -y(t)V\, dt$$

since it does not change if the functional is multiplied by a constant. Hence, the minimisation problem becomes

$$\min_{u(t)} \int_0^2 -y(t)\sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}\, dt = \min_{u(t)} \int_0^1 -y(x)\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

The functional that should be minimised is therefore

$$J(y) = \int_0^1 -y(x)\sqrt{1 + (y'(x))^2}\, dx$$

[5 marks]

b) The boundary conditions are

$$y(0) = 0, \qquad y(1) = 0$$

[2 marks]

Answers

c) The boat moves at a constant speed V and the total time of the trip is 2. Therefore, the total distance travelled by the boat is

$$\int_0^2 V\, dt = 2V = \text{ constant} = \int_0^2 \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}\, dt = \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Therefore, there is an integral length constraint in the problem, as follows

$$\int_0^1 \sqrt{1 + (y'(x))^2}\, dx = 2V$$

[4 marks]

d) The Lagrangian of the problem is

$$L = -y(x)\sqrt{1 + (y'(x))^2}$$

and the M function in the constraint functional is

$$M = \sqrt{1 + (y'(x))^2}$$

We define the augmented Lagrangian

$$L + \lambda^* M = -y(x)\sqrt{1 + (y'(x))^2} + \lambda^* \sqrt{1 + (y'(x))^2}$$

where λ^* is a constant Lagrange multiplier. Since the augmented Lagrangian does not depend explicitly on x, we can make use of the "no x" result of the Euler-Lagrange

equations as the first-order necessary condition for integral constraint optimality, as follows:

$$L + \lambda^* M - y'(x)\left(L + \lambda^* M\right)_{y'} = c \text{ with } c \in \mathbb{R}$$

Therefore, by substituting L and M in this condition:

$$-y(x)\sqrt{1+(y'(x))^2}+\lambda^*\sqrt{1+(y'(x))^2}-y'(x)\left(-\frac{y(x)y'(x)}{\sqrt{1+(y'(x))^2}}+\frac{\lambda^* y'(x)}{\sqrt{1+(y'(x))^2}}\right)=c$$

which yields

$$\frac{-y(x)+\lambda^*}{\sqrt{1+(y'(x))^2}}=c$$

This can be rearranged as

$$\left(\frac{dy}{dx}\right)^2=\left(\frac{-y(x)+\lambda^*}{c}\right)^2-1 \tag{13}$$

[6 marks]

e) The differential equation (13) can be solved by the method of separation of variables and can be shown to satisfy the solution

$$y(x)=-c\left(\frac{e^{\frac{x}{c}-a}+e^{-\frac{x}{c}+a}}{2}\right)+\lambda^*=-c\cosh(x/c-a)+\lambda^*$$

which is the extremal curve. The boundary conditions and length constraint functional can be used to find the values of the constant parameters c, a and λ^*. Note that, except from a sign difference, the form of the solution of this problem is the same as the catenary problem in the notes.

Answers

[3 marks]

Question 4

a) The Hamilton-Jacobi-Bellman (HJB) differential equation for this problem is

$$-V_t(t,x) = \inf_{u\in\mathbb{R}^m} \Big\{ (x - x_{ref}(t))^T Q(t)(x - x_{ref}(t)) + u^T R(t) u + \langle V_x(t,x), A(t)x + B(t)u \rangle \Big\}$$

[3 marks]

b) Since $R(t) > 0$, it is easy to see that the infimum of the quadratic function of u inside the infimum is a minimum and

$$u = -\frac{1}{2} R^{-1}(t) B^T(t) V_x(t,x) \tag{14}$$

Substituting the control (14) into the HJB equation, one obtains

$$\begin{aligned} -V_t(t,x) = {} & (x - x_{ref}(t))^T Q(t)(x - x_{ref}(t)) + (V_x(t,x))^T A(t) x \\ & - \frac{1}{4} (V_x(t,x))^T B(t) R^{-1}(t) B^T(t) V_x(t,x) \end{aligned}$$

[3 marks]

c) The boundary condition that must be satisfied by the value function is

$$V(t_f, x) = x^T M x - 2x^T M x_{ref}(t_f) + x_{ref}^T(t) M x_{ref}(t_f) \qquad \forall x \in \mathbb{R}^n \tag{15}$$

[3 marks]

Answers

d) i) In order for the assumed solution to match the boundary condition in part c), make

$$\begin{aligned} P(t_f) &= M \\ b(t_f) &= -M x_{ref}(t_f) \\ c(t_f) &= x_{ref}^T(t) M x_{ref}(t_f) \end{aligned}$$

[3 marks]

ii) By differentiating the assumed solution $V(t,x) = x^T P(t) x + 2x^T b(t) + c(t)$, one obtains

$$V_x(t,x) = 2P(t)x + 2b(t) \tag{16}$$

and

$$V_t(t,x) = x^T \dot{P}(t) x + 2x^T \dot{b}(t) + \dot{c}(t)$$

Then by substituting these terms in the HJB equation, yields

$$\begin{aligned} & x^T \Big(\dot{P}(t) + Q(t) + 2P(t)A(t) - P(t)B(t)R^{-1}(t)B^T(t)P(t) \Big) x \\ & + x^T \Big(2\dot{b}(t) - 2Q(t)x_{ref}(t) - 2P(t)B(t)R^{-1}(t)B^T b(t) + 2A^T(t)b(t) \Big) \\ & + \dot{c}(t) + x_{ref}^T(t)Q(t)x_{ref}(t) - b^T(t)B(t)R^{-1}(t)B^T b(t) = 0 \end{aligned}$$

Since this quadratic equation in x must be satisfied for all x, its coefficients need to be zero, and therefore the following system of ordinary differential equations for $P(t)$, $b(t)$, and $c(t)$ must be satisfied

$$\begin{aligned} \dot{P}(t) &= -P(t)A(t) - A^T(t)P(t) - Q(t) + P(t)B(t)R^{-1}(t)B^T(t)P(t) \\ \dot{b}(t) &= -\left(A^T(t) - P(t)B(t)R^{-1}(t)B^T \right) b(t) + Q(t)x_{ref}(t) \\ \dot{c}(t) &= -x_{ref}^T(t)Q(t)x_{ref}(t) + b^T(t)B(t)R^{-1}(t)B^T b(t) \end{aligned}$$

Notice that the solution for $P(t)$ is the same as the simpler LQR derivation in the notes (regulation instead of tracking problem). [6 marks]

iii) From (14) and (16) the optimal control law is

$$u^*(t) = -R^{-1}(t)B^T(t)P(t)x^*(t) - R^{-1}(t)B^T(t)b(t)$$

Note also that $c(t)$ has no effect on the control (even indirectly), and therefore it can often be ignored. [2 marks]

Answers